



WAYPOINT CONNECT

TESTING & VERIFICATION REPORT

Testing & Verification Report

What was tested, what passed, and what could not be verified

Project WaypointConnect — delivery planning and tracking for Waypoint Group's Fresh, Style and Tech brands

Event Tech-Triathlon 2026 · Team Dark Horse

Source Generated 2026-10-04 · repository commit c1e3eb6 + working tree

Accuracy rule: every statement is taken from the project files. Anything that could not be verified is marked "Not verified from the provided project files". No passwords or secrets are included.

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1. Testing strategy

1.1 Testing strategy

The project has one automated test suite: the planning engine and goods catalogue in `packages/shared/test` (Node test runner, 21 tests). Everything else is verified by typecheck, build, the official allocation checker, and end-to-end runs that the team recorded in `docs/ai-disclosure.md`. Those recorded runs are labelled **REPORTED** below: they are stated in the project files and were **not** re-executed for this report. Checks executed on **2026-10-04** while generating this report are labelled **PASS**.

Category	Present in repository?	Detail
Unit tests	Yes	21 tests in three files (engine 13, goods 3, schedule 5)
Integration / API tests	No test files	API flows were exercised by scripts during development (reported in the disclosure); the scripts are not stored in the repository
Frontend tests	No	No test files in <code>apps/web</code> ; screens were checked in headless Chromium (reported)
Type checking	Yes	<code>pnpm typecheck</code> (tsc in shared, api, web)
Build verification	Yes	<code>pnpm --filter @waypoint/web build</code> ; Docker image builds in CI
CI	Defined	<code>.github/workflows/ci.yml</code> : test, prisma generate, typecheck, check:2b, docker compose build. Run results are <i>Not verified from the provided project files.</i>
Docker / database	Reported	Full stack on real PostgreSQL in Docker and against the live URL (disclosure 2026-10-04). <code>docker compose config</code> validated on 2026-10-04

2. Executed checks

2.1 Executed on 2026-10-04 (Windows 11, Node v24.21.0, pnpm 9.12.0)

Test ID	Test	Expected result	Actual result	Status
V-01	<code>pnpm test</code>	All engine tests pass	tests 21, pass 21, fail 0, duration about 0.7 s	PASS
V-02	<code>pnpm typecheck</code> before <code>prisma generate</code>	No errors	apps/api: 11 TS errors (stale Prisma client: <code>driver</code> , <code>active</code> , <code>lastLoginAt</code> , role <code>admin</code>)	FINDING
V-03	<code>pnpm prisma generate</code> then <code>pnpm typecheck</code>	No errors in shared, api, web	All three workspaces: Done, exit code 0	PASS
V-04	<code>pnpm --filter @waypoint/web build</code> (via <code>apps/web build</code>)	tsc and Vite build succeed	134 modules transformed; JS 471.72 kB (135.27 kB gzip), CSS 46.43 kB; built in 4.25 s	PASS
V-05	Engine on the official peak day (<code>scripts/run-task2b.ts</code>)	Every order served or deferred with a reason	85 orders: 72 served on 32 trips, 13 deferred (all with a reason)	PASS
V-06	<code>docker compose config -q</code>	Compose file valid	Exit code 0	PASS
V-07	Live URL https://3-106-77-179.sslip.io	HTTPS reachable	HTTP 200	PASS
V-08	Official checker <code>check_allocation.py</code>	FEASIBILITY: PASSED	Not run: Python is not installed on the verification machine. README and disclosure state PASSED	REPORTED

2.2 Automated test inventory (all passed in V-01)

Test ID	File	Test	Expected	Status
U-01	engine.test.ts	trip time matches the booklet example: Fresh Gampaha, 2 rear docks + 1 street = 101 min	Pass	PASS
U-02	engine.test.ts	trip time matches the booklet example: Fresh Colombo, 4 street stops = 112 min	Pass	PASS
U-03	engine.test.ts	two Fresh trips of 101 + 112 min fit the 270 min window; a third trip is refused	Pass	PASS
U-04	engine.test.ts	chilled goods need a refrigerated vehicle	Pass	PASS
U-05	engine.test.ts	van-only outlets cannot be served by trucks	Pass	PASS
U-06	engine.test.ts	a vehicle only serves its home depot	Pass	PASS
U-07	engine.test.ts	vehicles in the workshop are refused	Pass	PASS
U-08	engine.test.ts	weight and volume caps are both enforced, with a readable reason	Pass	PASS
U-09	engine.test.ts	one brand and one district per trip	Pass	PASS

Test ID	File	Test	Expected	Status
U-10	engine.test.ts	Fresh trips must fit the 270 min pre-dawn window	Pass	PASS
U-11	engine.test.ts	weekly fuel quota is enforced when fuel use is known	Pass	PASS
U-12	engine.test.ts	priority: skipped-yesterday first, then Fresh chilled, Fresh ambient, Tech, Style	Pass	PASS
U-13	engine.test.ts	suggest plan on the official peak day (S1): every served trip is valid and every order has a decision	Pass	PASS
U-14	goods.test.ts	goods lines: packs always add up to the order units	Pass	PASS
U-15	goods.test.ts	goods lines are stable and follow the brand and temperature	Pass	PASS
U-16	goods.test.ts	store orders: weight and volume come from the pack, units are the packs	Pass	PASS
U-17	schedule.test.ts	ETAs follow the planning standard: outbound, then handling + inter-stop per stop	Pass	PASS
U-18	schedule.test.ts	a single Fresh trip waits so it reaches the first shop as it opens	Pass	PASS
U-19	schedule.test.ts	the second trip starts when the first ends (planning standard); the first leaves at 03:30	Pass	PASS
U-20	schedule.test.ts	stops are ordered by closing time and late arrivals are flagged	Pass	PASS
U-21	schedule.test.ts	S1 suggested plan: every trip gets a timetable that matches its trip minutes	Pass	PASS

3. Results recorded in the project files

3.1 Recorded in the project files (not re-executed here)

Test ID	Test	Expected result	Actual result (as recorded)	Status
R-01	Full stack on real PostgreSQL in Docker; every endpoint called over HTTPS and validated against <code>contract.ts</code>	All responses match the contract	Passed for dispatcher suggest/publish, loader claim/tick/problem/seal, driver sync incl. idempotent replay, POD, store order (ai-disclosure 2026-10-04)	REPORTED
R-02	Sign-in of all 188 accounts with own passwords; old logins rejected	Each account signs in with its own password	Recorded as passed. Conflicts with the source finding F-01 below.	REPORTED
R-03	Role isolation and depot lock	Dispatchers cannot open the other depot; non-admin blocked from admin routes	Recorded as passed (disclosure rows for personal accounts and admin console)	REPORTED
R-04	Admin actions kept after API restart	Assignments, switch-offs, resets and new accounts persist	Recorded as passed; screens checked in headless Chrome, no console errors	REPORTED
R-05	Next-run order cycle at 10:00 and 18:00 for both depots	Store order listed for the next run, then planned	Recorded as passed; next-day build tested twice (idempotent)	REPORTED
R-06	Offline driver flow: real offline reload via service worker; duplicate-safe resend	App opens offline; repeated event delivered once	Recorded as passed in Chromium (disclosure 2026-10-02)	REPORTED
R-07	Real-time refresh	Signal about 260 ms after an order; a new store order on an open dispatcher queue in 630 ms	Recorded as passed	REPORTED
R-08	Demo mode story rehearsed twice end to end	Every step passes; End demo returns to real time	Recorded as passed; plan identical each reset (OUT026 on VEH031 trip 2, 10 unavoidable deferrals)	REPORTED
R-09	Driver fingerprint unlock	Turn on, lock on reopen, wrong finger rejected, unlock, turn off	Recorded as passed with a virtual authenticator	REPORTED

4. Security, rules and gaps

4.1 Authentication and authorisation testing

Test ID	Test	Expected result	Actual result	Status
F-01	Source inspection of <code>POST /api/auth/login</code> (<code>apps/api/src/routes/auth.ts</code>)	Only the correct password for the account is accepted	The check <code>isDemoPw = ... password.trim().length > 0</code> is true for any non-empty password, so <code>valid</code> is always true for an existing account. Aliases (<code>dispatcher@</code> , <code>loader@</code> , <code>driver@</code> , <code>store@</code>) and a first-user-of-role fallback also exist. Not executed against a running server; the logic is direct from the code	FINDING
F-02	Source inspection of <code>requireRole</code>	401 without token, 401 for a switched-off account, 403 for another role	Implemented as expected in <code>auth.ts</code> (every route group uses it)	PASS
F-03	Source inspection: depot and outlet scoping	Dispatcher with depot locked; store routes scoped to <code>outletId</code> ; loader refuses other depot trips	Implemented in <code>dayOf()</code> , <code>ctxOf()</code> , <code>findTrip()</code>	PASS
F-04	Repository secrets	No password files in the repository	<code>data/user_credentials.csv</code> is tracked in git and contains per-account passwords. Its values are not reproduced in this report	FINDING

Recommended before relying on the system

- Remove the `|| password.trim().length > 0` operand and the alias / fallback logic from `auth.ts`.
- Remove `data/user_credentials.csv` from the repository and rotate `ACCOUNT_SECRET` / reset passwords, because its contents are in git history.
- Regenerate these documents afterwards so they describe the corrected behaviour.

4.2 Business-rule testing

Rule	Covered by	Status
Chilled needs reefer; van-only; home depot; workshop; weight/volume; one brand + district; fuel quota	U-04 to U-11	PASS
Trip minutes (booklet examples) and Fresh 270-minute window; max 2 trips	U-01 to U-03, U-10	PASS
Priority order	U-12	PASS
Suggest plan: every order has a decision; every served trip valid	U-13, U-21, V-05	PASS

Rule	Covered by	Status
Goods packs add up to units	U-14 to U-16	PASS
Timetable: ETAs, second trip start, closing-time order	U-17 to U-21	PASS
16:00 cutoff, current run, next operating day, Colombo time	No automated test; reported in R-05	REPORTED
Sealed / departed trips locked; second-deferral confirmation; driver one-vehicle rule; idempotent sync	No automated test in the repository; reported in R-01, R-04, R-06	REPORTED

4.3 Gaps

- No automated API, frontend or end-to-end tests are stored in the repository.
- CI results are *Not verified from the provided project files.*
- Docker build and run, and the official checker, were not re-executed for this report.